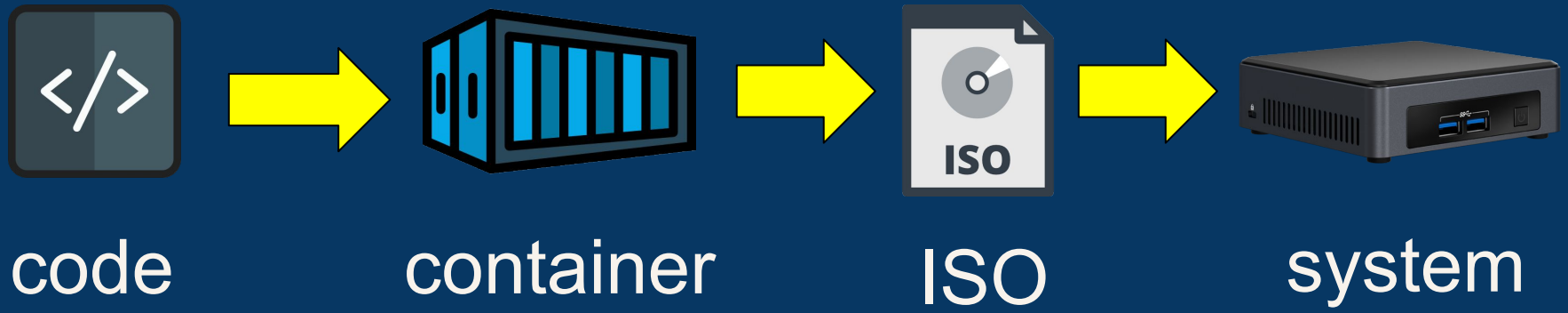
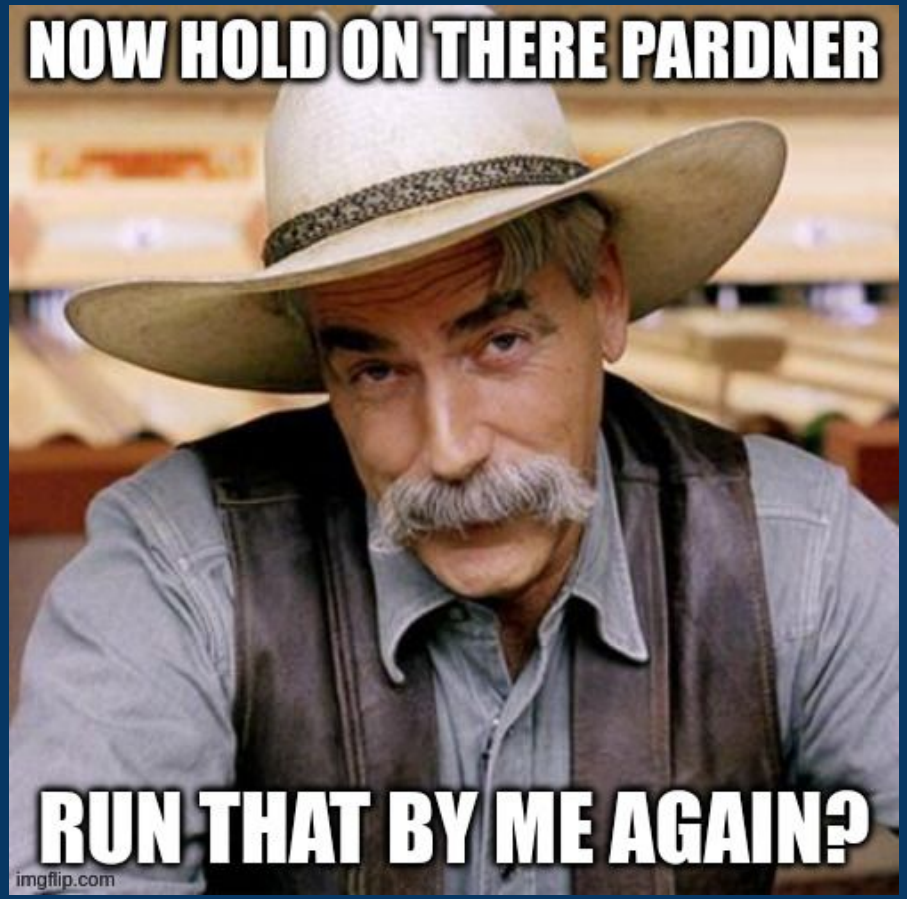
How to provision your hardware with … a container?

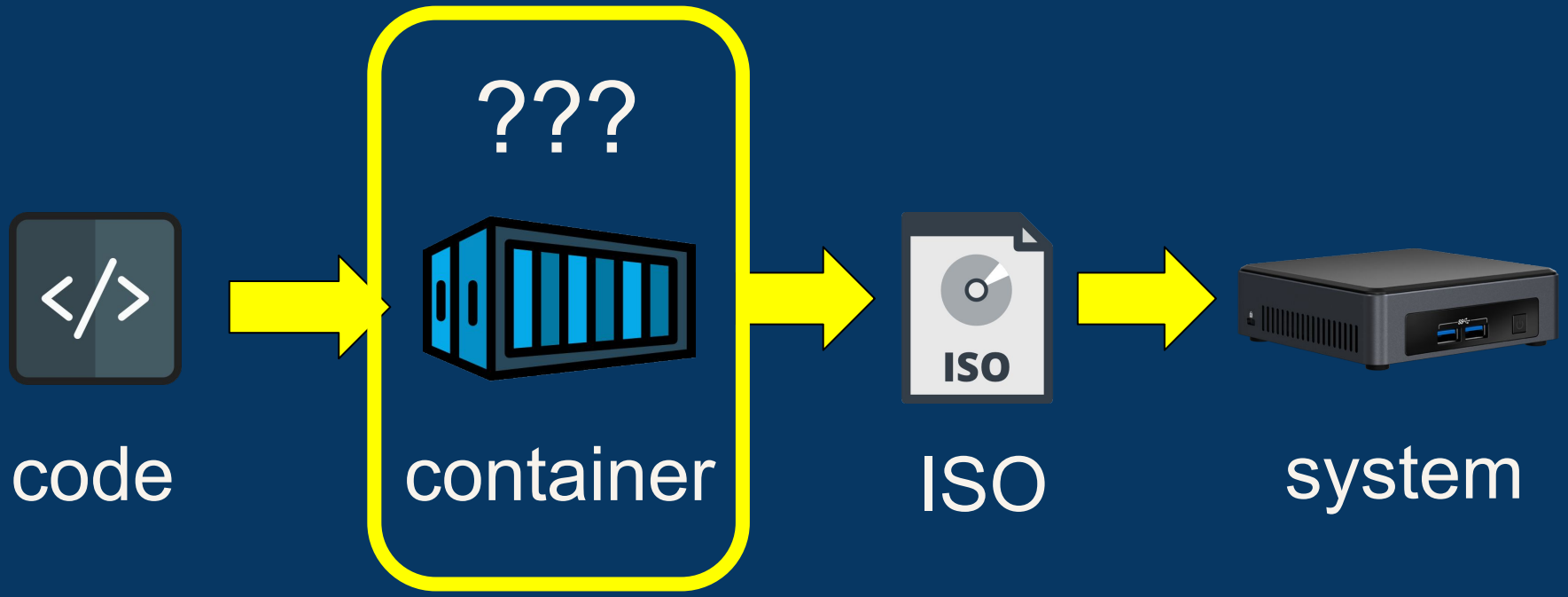
Josh Berkus

Red Hat

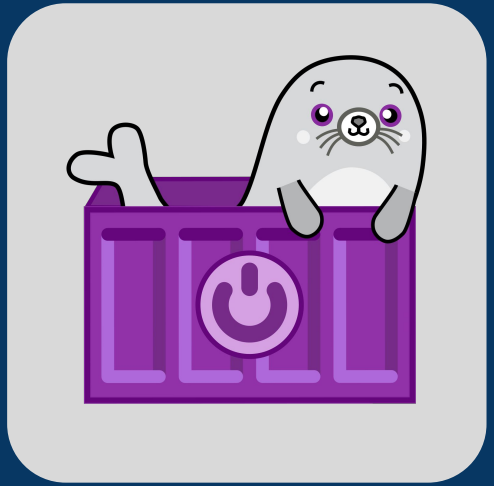
SCALE 23x


ISO
code
container
ISO
system


NOW HOLD ON THERE PARDNER
RUN THAT BY ME AGAIN?
imgflip.com


???
ISO
code
container
ISO
system

A brief digression into the history of OS deployment tech

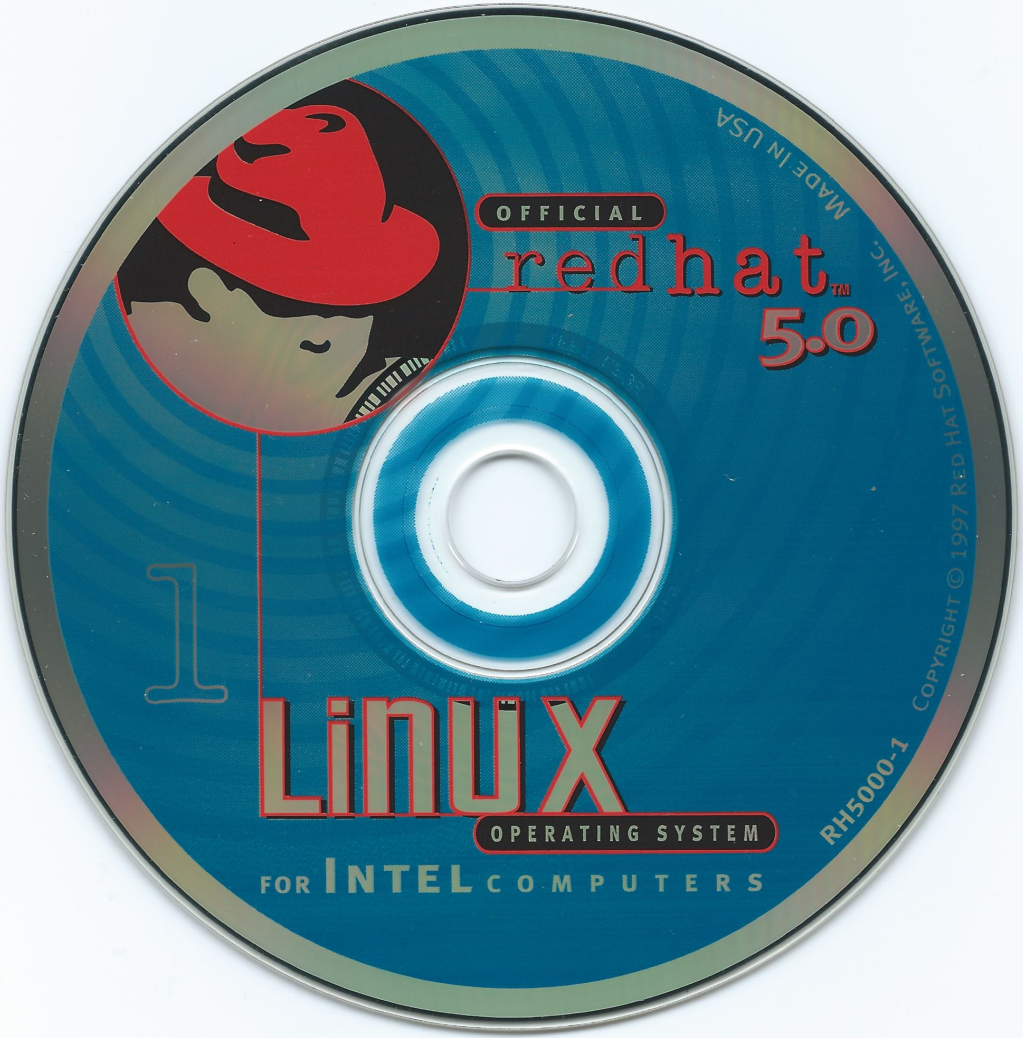


1995: Compile Everything

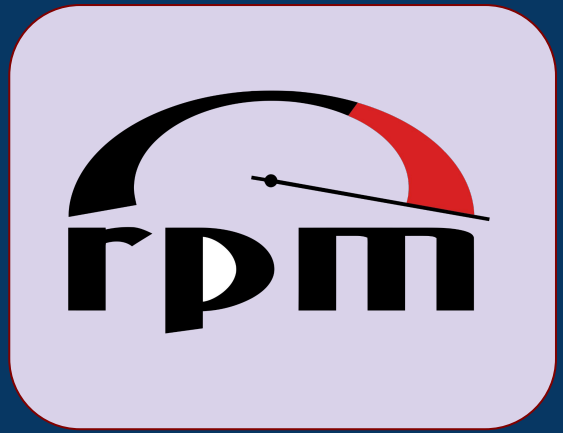
```
Linux Kernel Configuration
Arrow keys navigate the menu.  <Enter> selects submenus --->.
Highlighted letters are hotkeys.  Pressing <Y> includes, <N> excludes,
<M> modularizes features.  Press <Esc><Esc> to exit, <?> for Help.
Legend: [*] built-in  [ ] excluded  <M> module  < > module capable

    Code maturity level options  --->
    General setup  --->
    Loadable module support  --->
    Processor type and features  --->
    Power management options (ACPI, APM)  --->
    Bus options (PCI, PCMCIA, EISA, MCA, ISA)  --->
    Executable file formats  --->
    Device Drivers  --->
    File systems  --->
```

1998: Packages

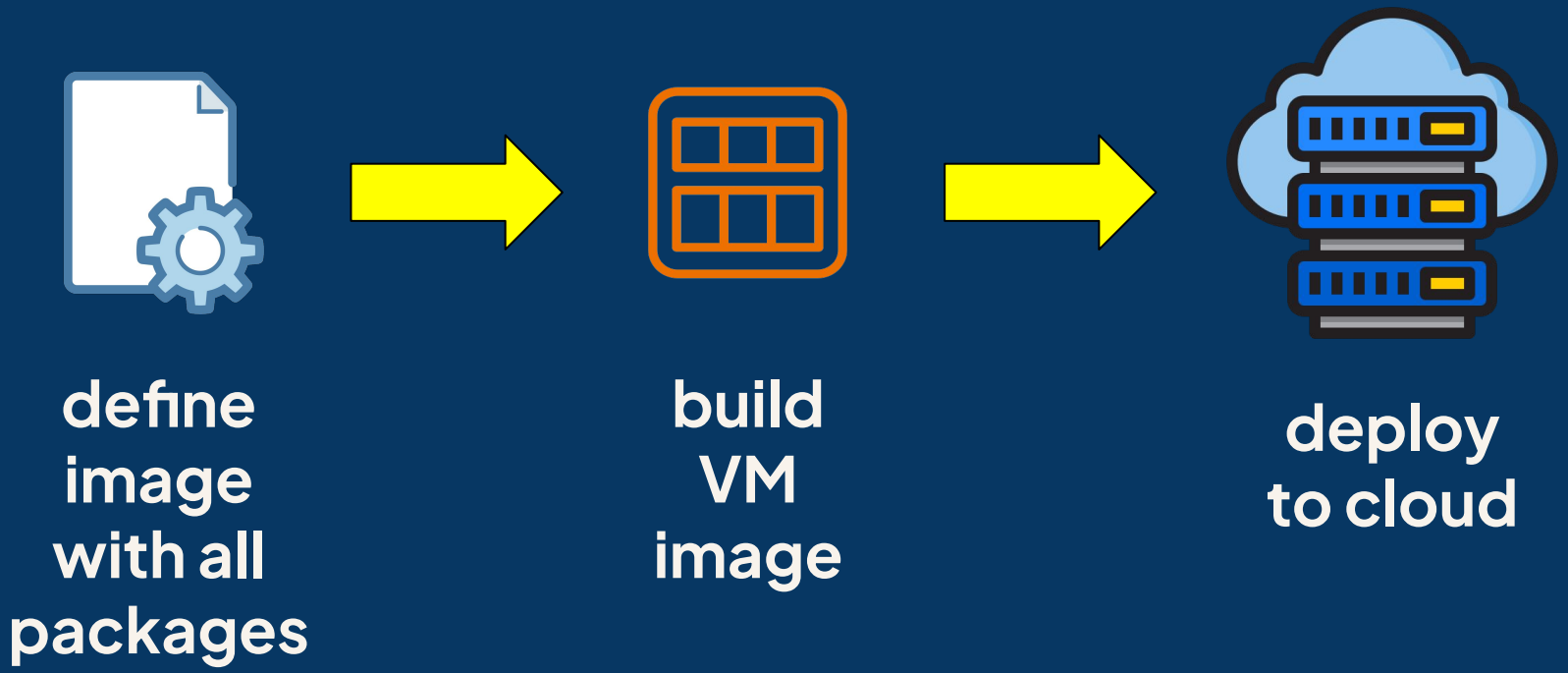


2005: Packages & ConfigMgmt



OS Install Scaling Problems

- Too slow for large numbers of machines
- Non-determinative outcomes
- Failures can't self-heal
- Hard to parallelize
- No rollbacks



2010: Deploying "Images"

VM Image Problems

Difficult composition tools

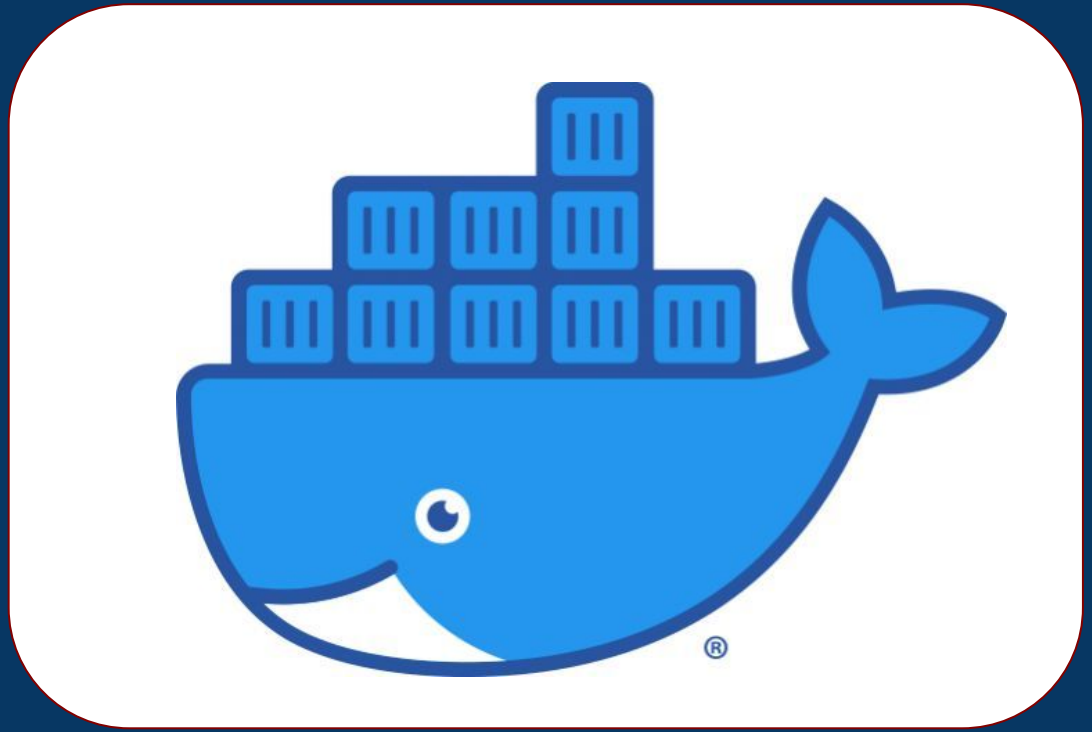
Images are slow to build

Large images are hard to distribute

Works poorly with bare metal

Still No Rollbacks

2014: Application Containers



Container Image Advantages

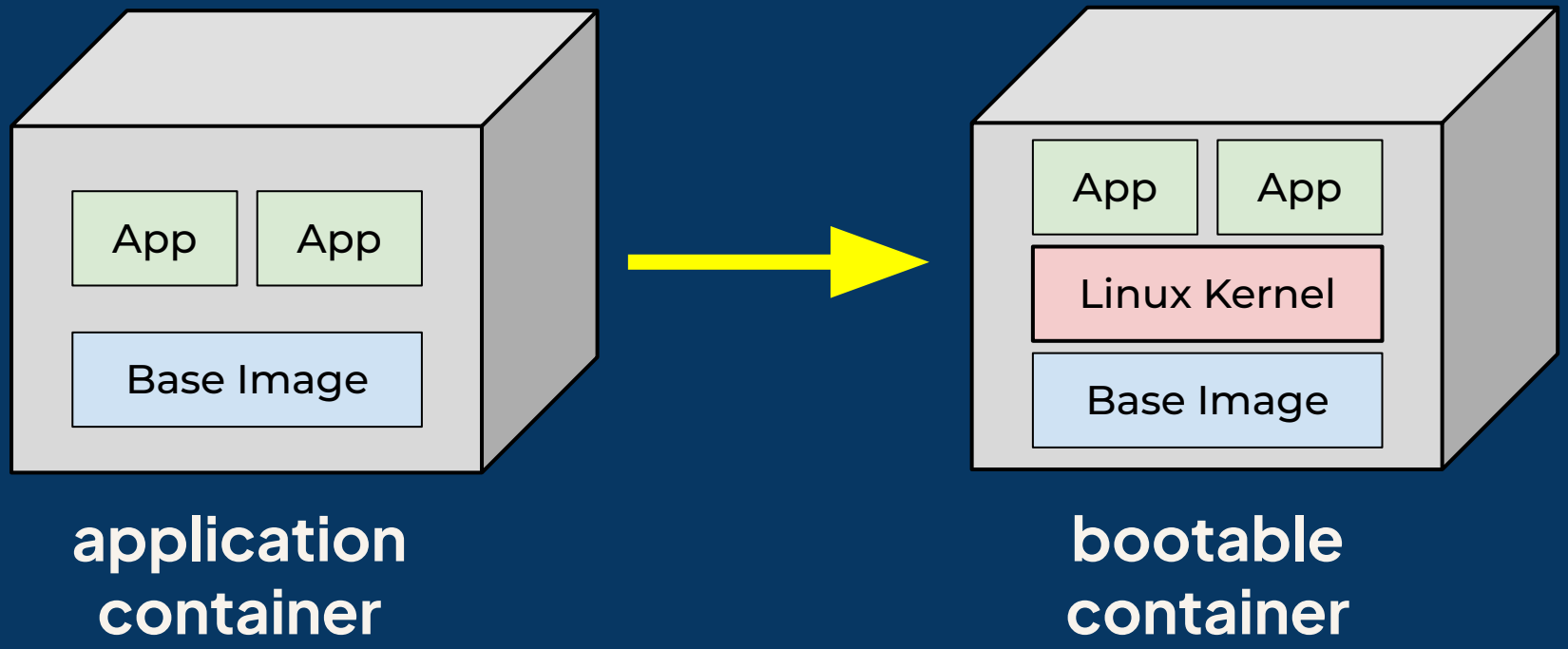
- Lots of good, familiar build tools
- Images build faster
- Smaller images in container registries
- Deploy on bare metal *and* cloud
- Rollbacks!

Introducing bootc

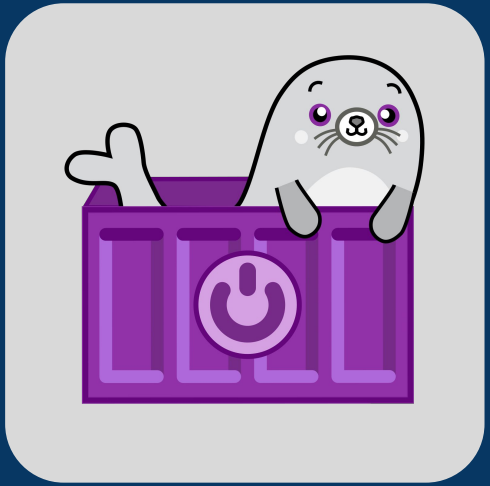
bootc: bootable containers

- builds bootable container images
- deploys containers as system images
- manages system status and updates
- supports rollback

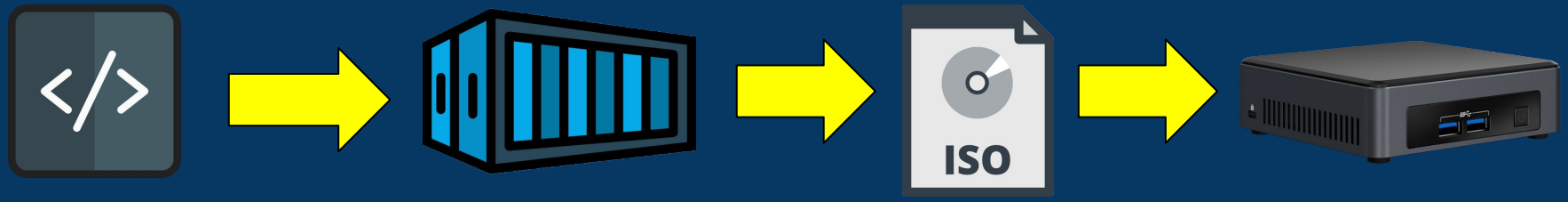
bootc: bootable containers



And now, back to building our system

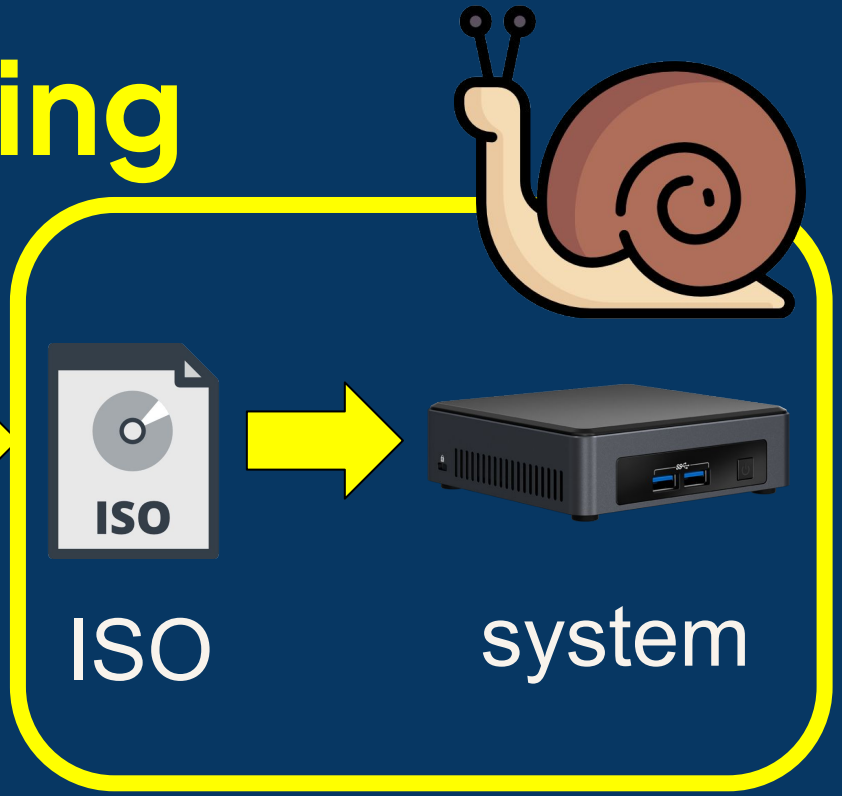


initial provisioning

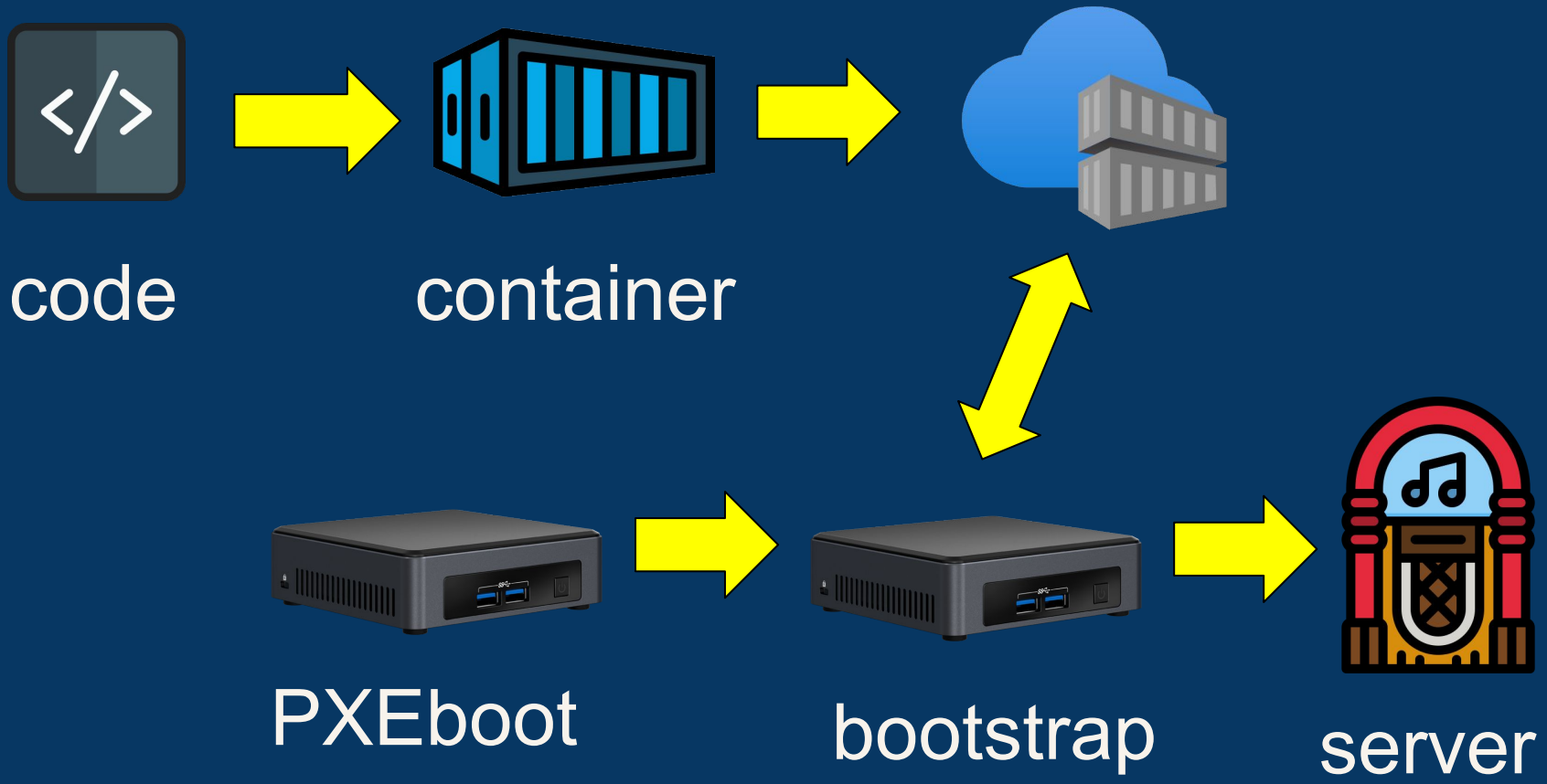


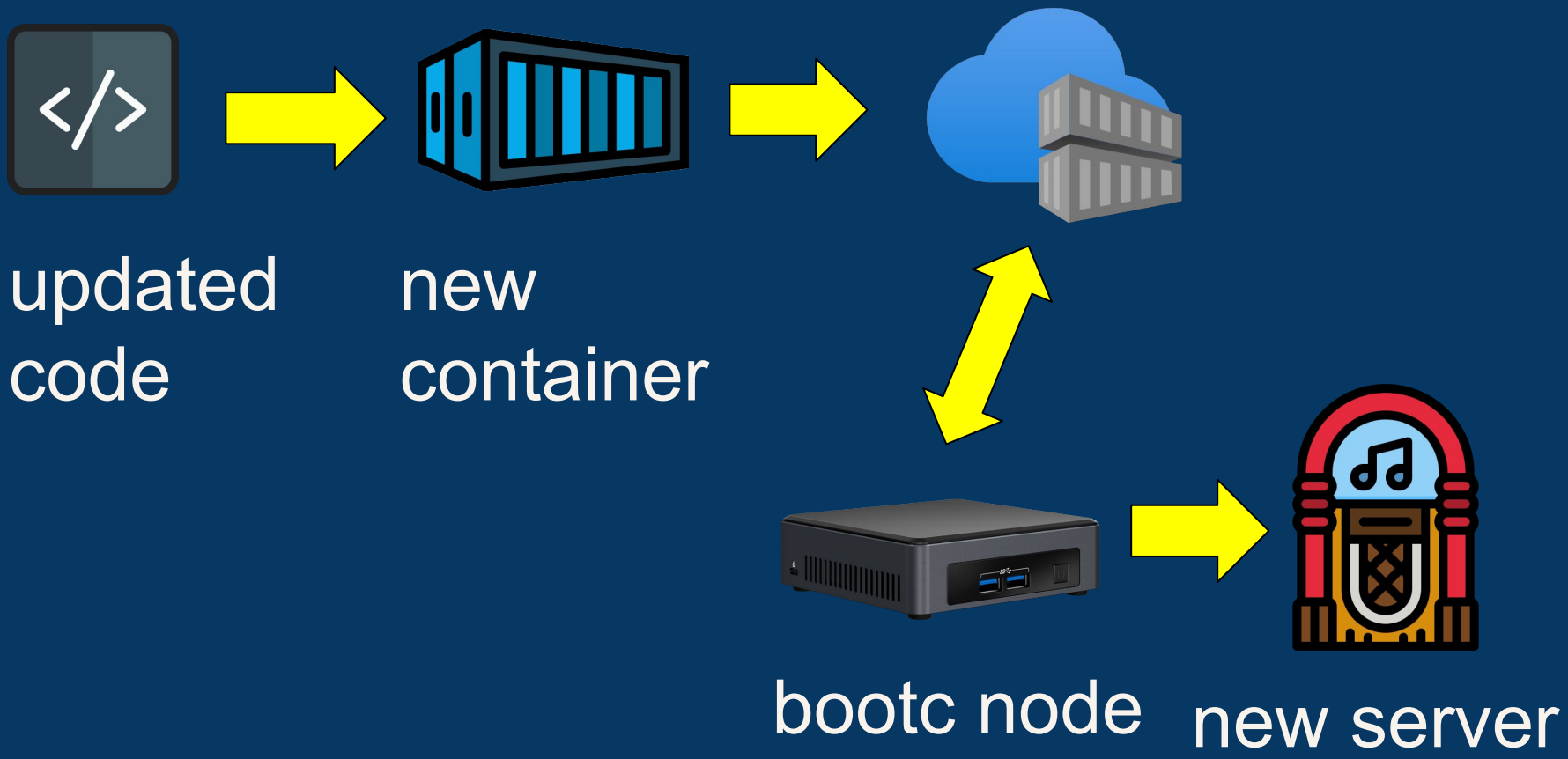
code container ISO system

initial provisioning



Checking storage configuration...
Installation
1) Ex1 Language settings
(Regional Setting Status)
2) Ex1 Keyboard Settings
(Keyboard Layout)
3) Ex1 Time settings
(Europe/Los_Angeles timezone)
4) Ex1 Network configuration
(Connected: eapinf)
5) Ex1 Disk partitioning
(Automatic partitioning selected)
6) Ex1 User creation
(No user will be created)
Progress
Setting up the installation environment
Configuring storage
Creating disklabel on /dev/sda
Creating sda1 on /dev/sda1
Creating sda2 on /dev/sda2
Creating sda3 on /dev/sda3
Creating sda4 on /dev/sda4
Running pre-installation scripts
Running pre-installation tasks
Installing the software
Begin/apt/ stating: /var/install/repo/installapp
WARNING
Beelink
intel
312-10112-01


code
container
PXEboot
bootstrap
server


updated code
new container
bootc node
new server

Demo code:

github.com/jberkus/bootc-device-demo

bootc update advantages

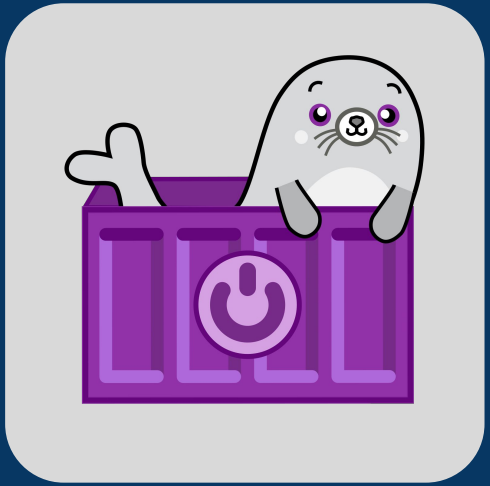
- No need to choose components
- Updates by efficient image layering
- No possibility of a partial update
- Rollback if update fails

Not Just For Edge Devices

fedora ATOMIC DESKTOPS

bazzite

What's next?



My jukebox distro

Generic bootc OS image

Quadlet autostart

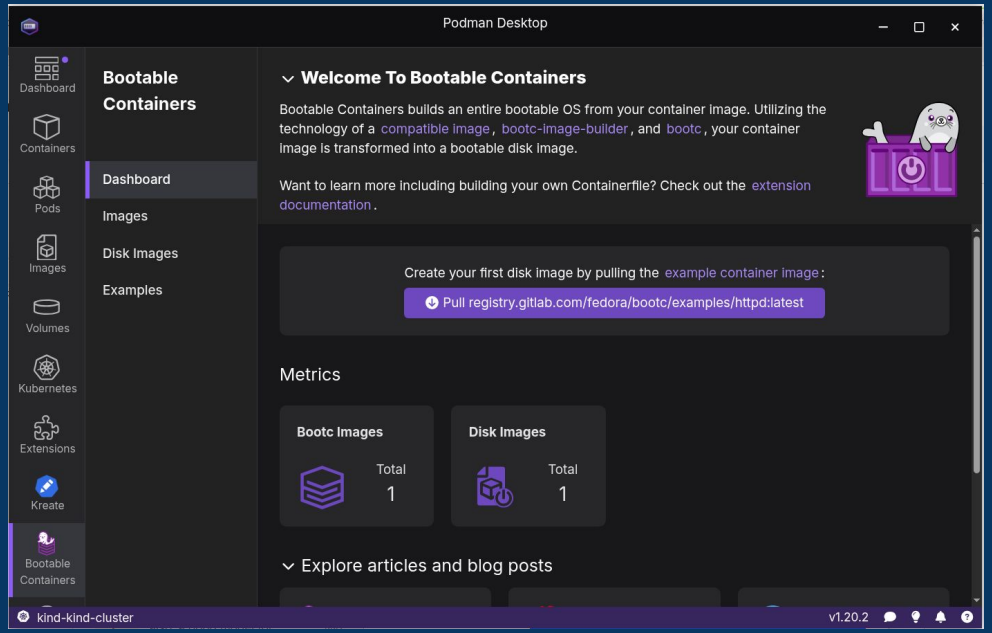
Preloaded containers

Media player: Plex? OSMC? Kodi?

Music storage?

Getting Started with bootc

1. Download **Podman Desktop** for your platform (Win/Mac/Lin)
2. Install the bootc extension
3. Play around with creating your own bootable images



Join the bootc community!

- bootc-dev.github.io
- Slack & Matrix chat, Friday dev meetings
- You can contribute!
  - Bug reports
  - Platform tests
  - Documentation

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